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From: http://118.185.187.137/moodle/mod/quiz/review.php?attempt=369217&cmid=3324

1-Number of Zeros in a Given ...
2-Majority Element
3-Finding Floor Value
4-Two Elements sum to x
5-Implementation of Quick Sort
Dynamic Programming
1-DP-Playing with Numbers
2-DP-Playing with chessboard
3-DP-Longest Common Subse...
4-DP-Longest non-decreasing ...
Competitive Programming
1-Finding Duplicates-O(n^2) Ti...
2-Finding Duplicates-O(n) Tim...
3-Print Intersection of 2 sorte...
4-Print Intersection of 2 sorte...
5-Pair with Difference-O(n^2)T...

3-DP-Longest Common Subsequence

Started on	Saturday, 12 September 2026, 2:31 PM
State	Finished
Completed on	Saturday, 12 September 2026, 2:36 PM
Time taken	5 mins 3 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct Mark 1.00 out of 1.00 Flag question

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1 a g **g t a b**

s2 **g** x **t** x **a** y **b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>

int main() {
    char s1[1001], s2[1001];
    if (scanf("%s %s", s1, s2) != 2) return 0;

    int len1 = strlen(s1), len2 = strlen(s2);
    int dp[1002][1002] = {0}; // Automatically zeroes out the array

    for (int i = 1; i <= len1; i++) {
        for (int j = 1; j <= len2; j++) {
            if (s1[i - 1] == s2[j - 1])
                dp[i][j] = dp[i - 1][j - 1] + 1;
            else
                dp[i][j] = (dp[i - 1][j] > dp[i][j - 1]) ? dp[i - 1][j] : dp[i][j - 1];
        }
    }
    printf("%d\n", dp[len1][len2]);
    return 0;
}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

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